WORKED SOLUTION

STEP II 2015, Question 7

Pure • Coordinate geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q7

The key idea

When two circles meet in two points, the line through those points (the common chord) is perpendicular to the line of centres. The circle D bisects C exactly when this chord is a diameter of C, so passes through the centre of C. Pythagoras in the right-angled triangle formed by the two centres and one meeting point then turns "bisects" into a single equation: (distance between centres)$^2 = r^2 - a^2$.

Solution

Part (i)

Let D have centre Q and radius r, and let O be the origin.

The general condition. Suppose D bisects C, meeting it at P and $P' = -P$, where $|P| = a$. Both points lie on D, so $QP = QP' = r$. Then Q is equidistant from P and P', so it lies on the perpendicular bisector of PP', which passes through O (the midpoint of PP'). Hence $OQ \perp OP$, and Pythagoras in triangle OQP gives

$$r^2 = OQ^2 + a^2. \qquad (*)$$

Conversely, if D is any circle with $OQ^2 = r^2 - a^2$ and $OQ > 0$, take P on C with $OP \perp OQ$. Then $QP^2 = OQ^2 + a^2 = r^2$, and likewise for $-P$, so D passes through two diametrically opposite points of C. Two different circles meet in at most two points, so these are the only meeting points and D bisects C.

Existence when $r > a$. Put the centre on the x-axis at distance $\sqrt{r^2 - a^2} > 0$ from O:

$$(x - \sqrt{r^2 - a^2}\,)^2 + y^2 = r^2.$$

Directly: subtracting $x^2 + y^2 = a^2$ from this equation gives $-2x\sqrt{r^2 - a^2} + (r^2 - a^2) = r^2 - a^2$, so $x = 0$ (as $r \neq a$), and then $y = \pm a$. The circles meet at exactly $(0, a)$ and $(0, -a)$, which are diametrically opposite on C. So this circle of radius r bisects C.

Impossibility when $r \leq a$. If some circle of radius r bisects C, then $(*)$ gives $r^2 = OQ^2 + a^2 \geq a^2$. So $r < a$ is impossible. If $r = a$, then $(*)$ forces $OQ = 0$, so D is C itself, which meets C in infinitely many points, not exactly two. Hence no circle of radius $r \leq a$ bisects C.

Part (ii)

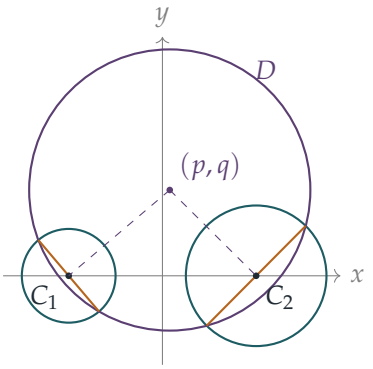
Let D have centre (p, q). Applying $(*)$ to each circle (with the origin shifted to the relevant centre),

$$(p+d)^2 + q^2 = r^2 - a_1^2,$$
$$(p-d)^2 + q^2 = r^2 - a_2^2.$$

Subtracting the second from the first, the q^2 and p^2 terms cancel, leaving $4pd = a_2^2 - a_1^2$, so

$$p = \frac{a_2^2 - a_1^2}{4d},$$

as required.



In the sketch ($d = 2$, $a_1 = 1$, $a_2 = 1.5$, $r = 3$) each common chord (amber) is a diameter of C_1 or C_2, perpendicular to the dashed line joining the centres.

For q, substitute p back into the first equation. First,

$$p + d = \frac{4d^2 + a_2^2 - a_1^2}{4d},$$

so

$$q^2 = r^2 - a_1^2 - \frac{(4d^2 + a_2^2 - a_1^2)^2}{16d^2} = \frac{16d^2r^2 - 16d^2a_1^2 - (4d^2 + a_2^2 - a_1^2)^2}{16d^2}.$$

The subtracted terms in the numerator expand to

$$\begin{aligned} 16d^2a_1^2 + (4d^2 + a_2^2 - a_1^2)^2 &= 16d^4 + 8d^2(a_2^2 - a_1^2) + 16d^2a_1^2 + (a_2^2 - a_1^2)^2 \\ &= 16d^4 + 8d^2(a_1^2 + a_2^2) + (a_2^2 - a_1^2)^2 \\ &= (4d^2 + (a_2 - a_1)^2)(4d^2 + (a_2 + a_1)^2), \end{aligned}$$

where the last line holds because $(a_2-a_1)^2+(a_2+a_1)^2=2(a_1^2+a_2^2)$ and $(a_2-a_1)^2(a_2+a_1)^2=(a_2^2-a_1^2)^2$. Therefore

$$q=\pm\frac{\sqrt{16r^2d^2-(4d^2+(a_2-a_1)^2)(4d^2+(a_2+a_1)^2)}}{4d}.$$

Ask yourself Where does the inequality come from? Nothing so far looks like an inequality.

The centre (p,q) is a real point, so $q^2\geq 0$. Since $16d^2>0$, this says

$$16r^2d^2\geq(4d^2+(a_2-a_1)^2)(4d^2+(a_2+a_1)^2),$$

which is the required condition on r.

We should also check that part (i)'s requirement $r>a_1$ (and $r>a_2$) is not an extra condition. Since $d>a_1$, we have $4d^2+a_2^2-a_1^2>0$, so $p+d>0$; then $r^2-a_1^2=(p+d)^2+q^2>0$. Similarly $d-p>0$ because $d>a_2$, so $r^2-a_2^2=(p-d)^2+q^2>0$. So the circles D found really do bisect both C_1 and C_2.

Checks

- Symmetric case $a_1=a_2=a$: then $p=0$, as symmetry about the y-axis demands, and the condition becomes $16r^2d^2\geq 4d^2(4d^2+4a^2)$, that is $r^2\geq d^2+a^2$. That is exactly $(*)$ with the centre at the origin, the closest possible centre to both $(\pm d,0)$.
- In the sketch's example, $p=0.15625$ and $q\approx 1.830$, and the intersection points of D with C_1 are approximately $(-1.353,-0.762)$ and $(-2.647,0.762)$, whose midpoint is $(-2,0)$, the centre of C_1. The smallest possible r there is about 2.377.
- Swapping C_1 and C_2 ($a_1\leftrightarrow a_2$, $d\to -d$) leaves p unchanged in form and leaves the inequality unchanged, as it should.

Common slips

- Forgetting the case $r=a$ in part (i): $(*)$ allows $OQ=0$, and you must say why $D=C$ does not count.
- Writing the condition as $OQ^2=r^2+a^2$, which confuses bisecting with meeting at right angles (orthogonal circles). Draw the triangle: the right angle is at the centre of C, so the hypotenuse is the radius r of D.
- Giving only $q=+\sqrt{\cdots}$: reflecting in the x-axis gives a second circle.

Beyond STEP

The quantity $(p-x_0)^2+(q-y_0)^2-r^2$ for a point (p,q) and a circle with centre (x_0,y_0) and radius r is called the *power* of the point with respect to the circle, and the set of

points with equal power with respect to two circles is a straight line, the *radical axis*; subtracting two circle equations, as we did, finds it. Here the bisecting condition says the power of the centre of C with respect to D is $-a^2$. These ideas, and the related inversion in a circle, are developed beautifully in Needham, *Visual Complex Analysis*, and many STEP-style uses appear in Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.